

Compiler infrastructure engineer with professional LLVM 22 and C#.NET static-analysis experience. Public compiler/runtime projects date to 2022, with CIL and native x86-64 code generation by 2023 and an AST → IR → x64 pipeline by 2024. I now build UniversalToolchain around deterministic compiler composition, with 1,324/1,324 tests passing.

2022: Compilers / VMs · 2023: CIL + native x86-64 · 2024–2025: AST / IR / backends → toolchain · 2026 → : Professional LLVM + static analysis

PROFESSIONAL EXPERIENCE		SKILLS
2026 — present	ISP RAS — Static Analysis Engineer Contribute to SharpChecker, ISP RAS's C#.NET static-analysis platform.	Compiler Infrastructure dependency/conflict resolution, component/provider selection, deterministic pass ordering, typed artifact routing, pre-execution checks
July — August 2026	MCST — Compiler Engineering Intern - Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety. - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.	Compiler / IR C++23, LLVM 22/IR, CFG/SSA, loop and interprocedural analysis, transformation applicability
Additional product experience — Diploma.school — client development: jQuery dropdown interactions, responsive mobile/desktop navigation, and UI regression fixes; confirmed no later than January 2025.		Backends / Runtime .NET, CIL/MSIL, native x86-64, SysV ABI, stack frames/calling conventions, register allocation, interpreter/runtime composition
		Verification regression suites, LLVM Verifier, interpreter/backend parity, differential execution
PROJECTS		EDUCATION
UniversalToolchain Language features resolve dependencies and conflicts, select capability/runtime providers, obtain deterministic contribution ordering, and build typed artifact routes before runtime materialization. Current manifest: 1,324/1,324 tests; dedicated checks cover interpreter/CIL parity and configuration integrity before execution.		HSE University — Software Engineering, 2026–2030
LLVM Interprocedural Global IV Optimization An LLVM 22 pass localizes supported induction-like globals after interprocedural effect analysis, affine evolution, and a separate applicability check; APInt models fixed-width wraparound. 29 positive/negative regression cases cover LLVM Verifier checks, structural expectations, idempotence, and observable before/after behavior.		RECOGNITION
x86-64 Compiler Backend & Register Allocation SSA/CFG validation → liveness/interference → register allocation/spills → phi lowering → SysV x86-64 emission. An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves; a separate public native-codegen lineage starts with Visk (2023) and Wist4 (2024).		LangDev'26 — Build the Language, Then Make the Abstractions Disappear Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.
		UniversalToolchain — Baltic Science and Engineering Competition First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.
		MEPhI Junior Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.
		Moscow / remote